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INTERIM FEDERAL SPECIFICATION

BURS, DENTAL EXCAVATING, TUNGSTEN CARBIDE

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal Agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this Interim Federal Specification.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers tungsten carbide excavating burs suitable for use in dental cavity preparations.

1.2 Classification.

1.2.1 Class. Excavating burs shall have shanks of the following classes, as specified:

Class AL	- Angle handpiece, latch.
Class AF	- Angle handpiece, friction grip.
Class AFS	- Angle handpiece, friction grip, short.
Class S	- Straight handpiece.

1.2.2 Shape and size number. Excavating burs shall have heads of the following shapes for the respective size number indicated:

Shape A - Round
Size Nos. - 1/4, 1/2, 2, 4, 4S, 6, and 8

Shape B - Inverted cone
Size Nos. - 33-1/2, 34, 35, 35S, 37, and 39

Shape C - Straight fissure, plain (noncrosscut) blade, flat end
Size No. - 56 and 57

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Shape D - Tapered fissure, plain blade, long head
Size Nos. - 169L and 170L

Shape E - Pear
Size No. - 330

Shape F - Straight fissure, crosscut blade, flat end
Size Nos. - 557, 558, 558S, and 559

Shape G - Tapered fissure, crosscut blade
Size Nos. - 699, 700, 701, 702, and 703

Shape H - End cutting
Size No. - 957

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

QQ-B-613	- Brass, Leaded and Nonleaded, Flat Products (Plate, Bar, Sheet, and Strip).
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

Federal Standards:

Fed.Std.No. 123 - Marking For Domestic Shipment (Civil Agencies)
Fed.Std.No. 151 - Metal; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547	- Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.
MIL-P-116	- Methods of Preservation - packaging.
MIL-Q-9858	- Quality Program Requirements.

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications, standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise stated, the issue in effect on date of invitation for bids or request for proposals shall apply.

American Society for Testing and Materials(ASTM):

D 5 - Test for Penetration of Bituminous Materials.

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3.1 Material.

3.1.1 Tungsten carbide. One-piece burs and the head of two-piece burs shall be fabricated from one solid piece of cemented tungsten carbide with cobalt binder. Titanium or tantalum carbide, or both, may be added to achieve the required properties.

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3.1.2 Steel. The shank, neck, and the base of the head of two-piece burs shall be fabricated from hardened and properly tempered carbon or alloy steel.

3.2 Bur classes, shapes and sizes. Bur classes, shapes and size numbers shall be as specified in 1.2.1 and 1.2.2.

3.2.1 Bur shank and overall length of bur. The shape and dimensions of the bur shank, and the overall length of the bur shall be as shown in table I for the class specified (see 1.2.1 and 6.1).

TABLE I. Shank shape and dimensions and overall bur length

Class	Shape and dimensions of shank	Overall length of bur, inch(es)
L	See Figure 1	0.875 ± 0.010
F	See Figure 2	0.755 ± 0.015
FS	See Figure 2	0.655 ± 0.015
	See Figure 3	1.750 ± 0.010

3.2.2 Bur head and neck. The bur head shall be as shown in Figure 4 for the indicated shape. The dimensions of the bur head and neck shall be in accordance with table II for the applicable shape and size number.

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TABLE II. Head and neck dimensions

Shape	Size number	Number of blades	Head diameter at largest place, ± 0.002 inch	Minimum length of carbide of head, inch	Taper or cone angle, degrees total /1	Maximum diameter of neck, inch
A	1/4	6	0.020	0.012	Not appl.	0.016
A	1/2	6	0.025	0.016	Not appl.	0.020
A	2	6	0.040	0.028	Not appl.	0.029
A	4	6	0.056	0.036	Not appl.	0.036
A	4S	6	0.056	0.036	Not appl.	0.036
A	6	8	0.072	0.053	Not appl.	0.044
A	8	8	0.091	0.074	Not appl.	0.050
B	33-½	6	0.025	0.016	6-16	0.020
B	34	6	0.032	0.022	6-16	0.025
B	35	6	0.040	0.024	6-16	0.028
B	35S	6	0.040	0.024	6-16	0.028
B	37	6	0.056	0.032	6-16	0.034
B	39	6	0.072	0.055	6-16	0.042
C	56	6	0.032	0.146	0-2	0.032
C	57	6	0.040	0.146	0-2	0.042
D	169L	6	0.036	0.193	2-6	0.035
D	170L	6	0.040	0.193	2-6	0.039
E	330	6	0.030	0.055	4-8	0.027
F	557	6	0.040	0.146	0-2	0.042
F	558	6	0.048	0.146	0-2	0.048
F	558S	6	0.048	0.146	0-2	0.048
F	559	6	0.056	0.168	0-2	0.054
G	699	6	0.036	0.146	4-8	0.034
G	700	6	0.040	0.146	4-8	0.039
G	701	6	0.048	0.146	4-8	0.047
G	702	6	0.064	0.168	4-8	0.057
G	703 /2	6	0.082	0.177	4-8	0.063
H	957	6	0.040	0.146	Not appl.	0.040

/1 The total for both sides of the angle.

/2 The 703 straight handpiece bur shall also conform to the following additional dimensions:

Distal head diameter: 0.058 ± 0.002 inch
Length of neck section: 5/16 ± 1/32 inch
Number of crosscut dentate grooves per blade: 5 to 7

3.3 Design and construction. The burs shall be suitable for use in the cavity preparation of natural dentition. The 703 straight handpiece bur shall also be used in surgical procedures. The bur shall consist of a shank, a neck, and a head. The blades shall be uniformly cut and sharpened. All burs shall be of two-piece construction except Class AF and AFS burs, which may also be of one-piece construction.

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.3.1 One-piece burs. One-piece burs shall be entirely fabricated from solid piece of cemented carbide as specified in 3.1.1.

.3.2 Two-piece burs. Two-piece burs shall have the shank, the neck, and base of the bur head fabricated from steel as specified in 3.1.2, and the d shall be fabricated from cemented carbide as specified in 3.1.1. The nt of two-piece burs shall be securely and permanently welded. The welded nt shall be machined so that the resultant shape has the appearance of an egral head unit.

.4 Runout. The total indicated runout of the neck of the burs ll not exceed 0.002 inch, when measured as specified in 4.3.2.

.5 Cutting ability. The maximum total time for cutting the brass sheet izontally shall be as specified in table III for the applicable bur size ber, when tested as specified in 4.4.2.

.6 Durability. The minimum total depth of the 25 holes shall be as speci- d in table IV for the applicable bur size number, when tested as specified 4.4.3.

.7 Strength of neck and joint. The neck shall not break on any of the s, and the joint on two-piece burs shall not fracture, when tested as cified in 4.4.4. The nondamaging test load shall be as specified in table for the applicable bur size number.

.7.1 Increase in runout. The increase in runout shall be no greater than 03 inch, when determined as specified in 4.4.4.3.

.8 Finish. The bur head shall have a finely honed finish with sharpened des. The neck and shank of the burs shall be smooth. The steel portion two-piece burs shall be nickel or chromium plated.

.8.1 Autoclaving (applicable to all two-piece burs). The bur shall show corrosion, after having been subjected to the autoclave test as specified 4.4.5, except that corrosion present at the welded joint shall be disre- ded, and shall not be cause for rejection.

.8.2 Surface roughness (applicable only to class AF and AFS burs). The face roughness of the shanks of class AF and AFS burs shall be no more n the root-mean-square (RMS) average deviation of 20 microinches in the gitudinal direction of the shank, when determined as specified in 4.4.6.

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TABLE III. Cutting ability requirements

Bur size number	Thickness of brass sheet, inch	Depth of /1 setting for cut, inch	Projection of bur tip through sheet, inch	Maximum total cutting time, seconds
1/4	0.011	0.015	0.004	40
1/2	0.011	0.015	0.004	40
2	0.011	0.022	0.011	25
4	0.020	0.034	0.014	25
4S	0.020	0.034	0.014	25
6	0.020	0.042	0.022	25
8	0.020	0.052	0.032	25
33-½	0.011	0.016	0.005	25
34	0.011	0.016	0.005	25
35	0.011	0.016	0.005	25
35S	0.011	0.016	0.005	25
37	0.020	0.025	0.005	50
39	0.020	0.025	0.005	50
56	0.020	0.060	0.040	50 wet /2
57	0.020	0.060	0.040	50
169L	0.020	0.060	0.040	50 (wet) /2
170L	0.020	0.060	0.040	50 (wet) /2
330	0.011	0.016	0.005	25
557	0.020	0.060	0.040	50
558	0.020	0.060	0.040	50
558S	0.020	0.060	0.040	50
559	0.020	0.060	0.040	50
699	0.011	0.051	0.040	50
700	0.020	0.060	0.040	50
701	0.020	0.060	0.040	50
702	0.020	0.060	0.040	50
703	0.020	0.060	0.040	50
957	(Tests not applicable)			

/1 The "Depth of setting for cut" indicates the depth of the bur tip below the top surface of the brass sheet. The difference between this depth and the "Thickness of brass sheet" is the "Projection of the bur tip through sheet" (distance from the bottom surface of the brass sheet downward to the tip of the bur).

/2 A lubricant or other suitable cutting fluid shall be used for the cutting test on the 56, 169L and 170L burs only.

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TABLE IV. Durability and strength requirements

ır size umber	Minimum total depth of 25 holes, inch	Nondamaging test load, pounds
/4	0.200	2.0
/2	0.240	3.0
	0.400	3.5
	0.400	5.5
S	0.400	5.5
	0.400	7.0
	0.400	7.0
3-½	0.240	3.0
4	0.240	3.0
5	0.300	3.5
5S	0.300	3.5
7	0.400	5.5
9	0.400	7.0
6	0.325	3.0
7	0.400	4.0
69L	0.300	2.0
70L	0.325	2.0
30	0.300	3.0
57	0.400	4.0
58	0.400	5.0
58S	0.400	5.0
59	0.400	7.0
99	0.300	3.5
00	0.300	3.5
01	0.400	5.5
02	0.400	7.5
03	0.400	9.0
57	Not applicable	4.0

3.9 Marking. Marking of burs shall be optional. The use of a trademark acceptable. The type of marking or placement of the marking shall not versely affect or interfere with fit, function and operation of the bur th handpiece. The marking shall be distinguishable to the unaided eye.

3.10 Instructions. Adequate and accurate instructions shall be furnished th each unit of issue on the use and care of the burs. Instructions shall clude methods of sterilization and sanitization. The instructions shall so include the type of solutions, both suitable and unsuitable for cold nitization of the burs.

3.11 Workmanship. Workmanship shall be first class throughout. The burs all be free from defects which detract from their appearance or may impair eir serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or materials, tungsten carbide and steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Inspection lot. An inspection lot, for purposes of sampling and testing in accordance with MIL-STD-105, shall consist of no more than 10,000 burs of the same size, manufactured under essentially the same conditions. Each unit package of burs shall be permanently marked with a serial number, or a combination of letters and numbers, which shall refer to the manufacturer's records for that particular lot or batch.

4.2 Quality conformance inspection. The examinations required to assure conformance of those burs or lots of burs to be offered for acceptance are classified as specified in tables V, VII and VIII (see 4.2.1, 4.3 and 4.3.3.1) and 4.3.3.2. The tests required for acceptance are specified in table VI of 4.2.2.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as shown in table V. The unit of product for examination purposes shall be the individual bur.

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TABLE V. Examination criteria

pe defect	Inspection level	AQL(Percent defective)
jor	II	1.5
nor	II	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance th MIL-STD-105 and as specified in table VI. The acceptance number shall be ro for all sample sizes. The unit of product for test purposes shall be as ated in the table.

TABLE VI. Test criteria

aracteristic	Requirement	Test procedure	Inspection level	Unit of product
emical mposition	3.1	4.4.1	(See 4.2.3)	Raw material
tting ility	3.5	4.4.2	S-1	5 burs
rability	3.6	4.4.3	S-1 /1	5 burs
ck and joint rength	3.7	4.4.4	S-3	1 bur
crease in nout	3.7.1	4.4.4.3	S-3	1 bur
toclaving	3.8.1	4.4.5	S-1	1 (two piece bur)
rface ughness	3.8.2	4.4.6	S-1	1 (class AF and AFS burs)

If the supplier has a quality program in conformance with MIL-Q-9858, five burs manufactured from each lot of raw carbide material shall be tested.

4.2.3 Raw material. A sufficient quantity of each lot of raw material ed in the manufacture of the burs shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same burs shall be used for e determination of two or more test characteristics.

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4.3 Examination. The burs shall be examined for the possible defects listed in table VII. The examination shall not be limited to those defects shown in the table. Unless otherwise indicated in the table, the defect listed shall apply to all classes, shapes, and size numbers of tungsten carbide excavating burs covered by this specification.

TABLE VII. Classification of defects

Category	Defect
Major:	
101	Bur size number or head shape not as specified.
102	Bur shank not class specified.
103	Total indicated runout exceeds 0.002 inch.
104	Head diameter exceeds specification requirements at maximum cross-section.
105	Diameter of shank exceeds specification limits.
106	Length of carbide of head less than minimum requirement for applicable bur size number.
107	Width of groove at socket end of shank less than 0.0335 inch (class AL burs).
108	Shoulder-to-shoulder measurement on shank less than 0.0709 inch (class AL burs).
109	D-flat dimension on shank exceeds 0.0709 inch (class AL burs).
110	Diameter of socket groove on shank exceeds 0.0571 inch (class AL burs).
Minor:	
201	Cutoff end on shank not as shown in applicable figure.
202	Burs not free from defects such as chipped blades, cracks, or pits.
203	Number of blades not as specified for the applicable bur size number.
204	Finish on bur not as specified.
205	Overall length of bur not within specification limits.
206	Maximum neck diameter exceeds specification limits.
207	Shank dimensions, other than those specified under major defects, not within specification limits.
208	Taper or cone angle of head (when applicable) not within specification limits.
209	Instructions for use inaccurate, inadequate, or missing.
210	Welded joint of two-piece bur does not give the appearance of an integral head unit.
211	Material marking on shank either missing or faint and not readily visible to the unaided eye. (Classes AL and S burs).
212	Packaging and packing not as specified.
213	Marking on packaging or pack incorrect, illegible, incomplete, or missing altogether.

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4.3.1 Dimensions. Examination of the dimensions, as specified in 3.2, shall made with a toolmaker's microscope, or other instrument of equal accuracy. nufacturer's gages may be used if properly calibrated checking standards e used in conjunction with the gages.

4.3.2 Runout. The bur shall be placed and held in a "V" block or pre- ision chuck, and rotated slowly. A dial indicator sensitive to 0.0002 inch and uipped with a knife-edge foot, or a shadow-graph or other suitable equipment f equal accuracy, shall be employed. If a dial indicator is used, the indi- tor shall be mounted with the shaft perpendicular to the axis of the bur, d so positioned that the foot contacts the bur neck at the point of smallest iameter. The maximum and minimum readings obtained during one complete 360- egree rotation of the bur shall be measured and recorded to the nearest 0.0005 nch. The difference between these two readings shall be taken as the total dicated runout.

4.3.3 Visual examination. Examination for possible bur defects not per- ptible to the unaided eye shall be made using a 10X magnification.

4.3.3.1 Examination of preparation for delivery. An examination shall be de to determine compliance with the packaging, packing and marking require- nts of section 5. Defects shall be scored as specified in table VIII. mpling shall be in accordance with MIL-STD-105. The sample unit shall be one ntainer fully prepared for delivery. The lot shall be the number of con- iners offered for inspection at one time. The inspection level shall be S-2 th AQL of 4.0 expressed in terms of percent defective.

TABLE VIII. Examination of preparation for delivery

amine	Defect
ntents	not as specified.
ntainer	not as specified.
rking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
terial	Component missing or damaged.
rkmanship	Bulging or distoration of containers.

4.3.3.2 Examination of closure, waterproofing and banding of container. When e shipping container is required to comply with PPP-B-636, the examination r defects in closure, waterproofing or banding shall be in accordance with e appendix of the box specification.

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4.4 Test procedures.

4.4.1 Chemical composition. In the absence of certificates of quality for the raw material, the samples, selected as specified in 4.2.3, shall be analyzed in accordance with Method 111 or 112 of Fed. Std. No. 151. In case of dispute, Method 111 shall be used for referee analysis. (Separate analyses shall be made on the carbide and the non-carbide portions of two-piece burs.)

4.4.2 Cutting ability.

4.4.2.1 Test material. The test material shall be cartridge brass sheet conforming to chemical composition 2 of QQ-B-613. The sheet shall have a Rockwell hardness of B40 to B65 and shall have a thickness of 0.011 inch (24 gage Brown and Sharpe) or 0.020 inch (29 gage B & S), as specified in table III. The sheet shall be cut into strips of suitable size for use in the test equipment as specified in 4.4.2.2.

4.4.2.2 Equipment for tests. The equipment shall consist of a precision chuck capable of holding the burs to be tested, a movable carriage for the test material, and the necessary accessories for measuring within 0.5 second the time required for the bur to cut 0.020 inch. The equipment shall be capable of applying a constant one-pound contact load between the bur and the test strip perpendicular to the bur axis.

4.4.2.3 Test procedure. A test bur and a brass sheet of the proper thickness shall be placed in the test equipment. The bur shall be positioned perpendicular to the brass strip and the relative heights adjusted as specified in table III. The bur shall be rotated at a speed of 4350 $\pm$ 50 revolutions per minute (r.p.m.) and brought into contact with the edge of the brass strip. A preliminary cut equal to one-half to one bur diameter shall be made before readings are taken. The time required for the bur to cut horizontally a distance of 0.020 inch into the sheet shall be determined to $\pm$ 0.5 second. The test shall be repeated until a total of 5 burs have been tested.

4.4.3 Durability.

4.4.3.1 Test material. The test material shall be crystal plate glass, a minimum of 1/8 inch thick, having a Knoop hardness number of 450 to 510 with a one kilogram load.

4.4.3.2 Equipment for test. The equipment shall consist of a true-running chuck capable of holding and rotating the bur being tested, and a movable platform or carriage suitable for carrying the flat surface of a glass test block into contact with the bur at an angle of 22-1/2 $\pm$ 1/2 degrees from the axis of the bur. The equipment shall be capable of maintaining a constant contact load of one pound perpendicular to the glass surface.

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4.4.3.3 Test procedure. A test bur shall be placed in the chuck of the st equipment and rotated at 1550 ± 50 r.p.m. The glass test plate shall be owly brought into contact with the bur, and held in contact with it. The r shall be permitted to cut into the glass, with turpentine used as the tting fluid. After 2000 revolutions of the bur the plate shall be moved to pose a fresh, uncut portion of flat glass surface, and the test shall be re- ated until a total of 5 holes have been drilled by the test bur. The bur all then be removed, and a second test bur inserted into the chuck. The st shall be continued until a total of 5 burs have been tested, and a total 25 holes drilled. The plate shall be removed and the depth of holes shall measured with a sharp point on a dial gage sensitive to 0.0002 inch. (The int on the gage shall meet the requirements of ASTM Designation D 5).

4.4.4 Neck and joint strength.

4.4.4.1 Equipment for test. The equipment shall consist of a true-running uck suitable for holding and rotating the test bur, and a loading device pable of applying the load indicated in table IV through an anvil with a flat rface at an angle of 22-1/2 ± 1/2 degrees to the bur axis. The chuck shall ip at least a 3/8-inch portion of the bur shank starting at the intersection the neck and shank.

4.4.4.2 Test procedure. Following an initial examination for total indi- ted runout as specified in 4.3.2, those burs which have been selected for e neck and joint strength test shall be rotated in the chuck to that posi- on which shows the highest deflection reading on the dial indicator gage. load of the magnitude specified in table IV shall be slowly applied at an gle to the bur axis of 22-1/2 ± 1/2 degrees, in the direction which tends further increase the dial deflection reading. The neck and joint of the r shall be observed for evidence of breakage. Should breakage not have curred at the point of maximum deflection, the load shall be removed and e examination for increase in runout determined as specified in 4.4.4.3.

4.4.4.3 Increase in runout. The increase in runout shall be determined ly on those samples which have been subjected to the neck and joint strength, d shall be performed in the manner already specified in 4.3.2 for runout. e difference between the final and the initial indicated runout shall be ken as the increase in runout.

4.4.5 Autoclaving. The test burs shall be cleaned in accordance with the structions supplied by the manufacturer. The burs shall then be autoclaved 270 ± 5 degrees Fahrenheit (F.) and 27 ± 1 pounds per square inch pressure r a minimum of 20 minutes. After the burs are removed from the autoclave, e entire surface area of the non-carbide portion of the bur shall be examined r evidence of corrosion.

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4.4.6 Surface roughness. The surface roughness of class AF and AFS burs shall be determined by using a Brush Surfindicator, or other equipment of equal accuracy. The RMS average shall be determined by squaring the height measurements of the peaks and valleys of the surface of the shank, obtaining the averages of the squared values, and then extracting the square root of this average square.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C, as specified (see 6.1).

5.1.1 Unit of issue. (Military only) One package consisting of 10 burs, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.2.2 Preservative. Not required.

5.1.2.3 Unit package. Ten burs of one size only shall be individually blister packaged in strip form. The strip shall be perforated in such a manner to permit removal of burs one at a time and to permit folding the strip into a compact package. The folded strip shall be inserted into a snug fitting, clear plastic box having a hinged cover and friction grip closure. As an alternate, each bur shall be individually packaged in a suitable plastic holder with a cover. Ten (10) individually packaged burs shall be further packaged in a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676 and overwrapped with moisture resistant cellulose.

5.1.2.4 Intermediate package. Ten (10) unit packages shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidantal opening.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. One hundred (100) intermediate packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification. In addition, fiberboard boxes shall conform to the special requirements as specified in PPP-B-636.

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5.2.1.2 Case liner. Each level A wood box shall be lined with a water-roof case liner conforming to MIL-L-10547. Closure and sealing shall con-orm to applicable paragraphs of appendix thereto. Case liner shall not be equired for fiberboard boxes. Each fiberboard box shall be waterproofed n accordance with 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. One hundred (100) intermediate packages shall e packed in an exterior container designed for a type 2 load and constructed n accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as speci-ied in the appendix of the applicable box specification. Fiberboard box shall e closed by method II of PPP-B-636. In addition, fiberboard boxes shall con-orm to the special requirements as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial com-ercial containers of the type, size, and kind commonly used for the purpose, o constructed as to insure acceptance and safe delivery by common or other arriers, at the lowest rate, to point of delivery called for in the contract r purchase order.

5.3 Marking.

5.3.1 Civil agencies. Unless otherwise specified, marking for shipment hall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package and each intermediate package hall be marked in accordance with MIL-STD-129. Contract or purchase order umber and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be in accordance ith MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options per-itted herein and include the following information in procurement:

(a) Title, number and date of this specification.
(b) Shank class and bur size number.
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

Preparing activity:
DSA-DM

Project No. 6520-1427

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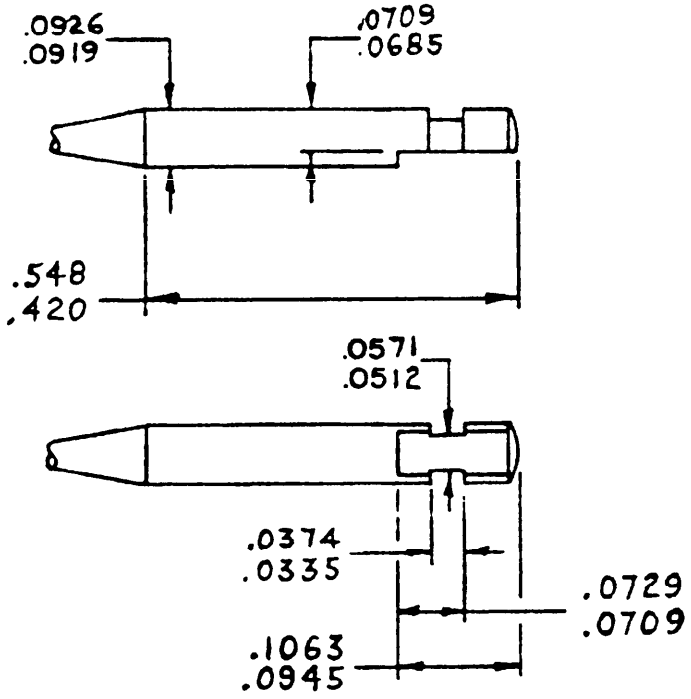


FIGURE 1. CLASS AL

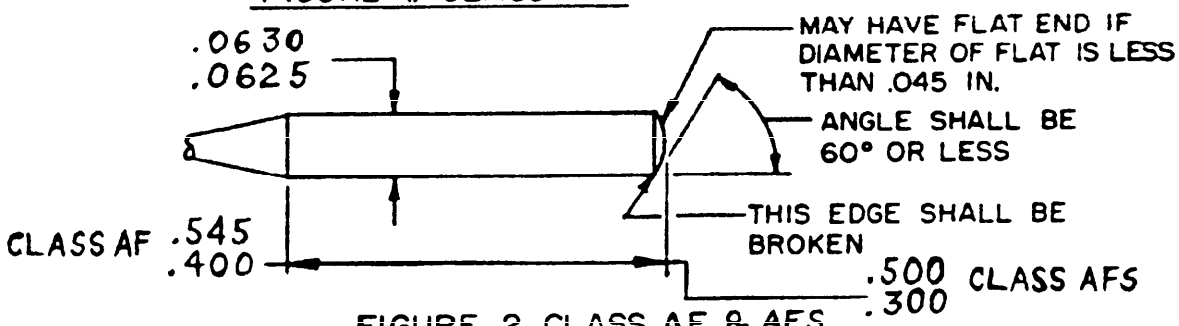


FIGURE 2. CLASS AF & AFS

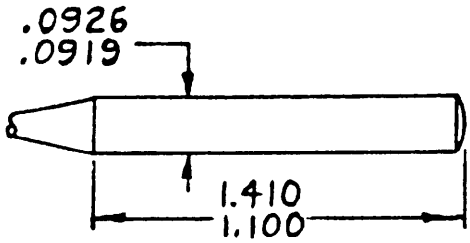
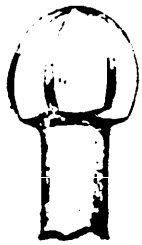


FIGURE 3. CLASS S

NOTE:
ALL DIMENSIONS ARE IN INCHES

DIMENSIONS OF SHANK END OF BUR

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Shape A

ROUND



Shape B

INVERTED CONE



Shape C

STRAIGHT FISSURE,
PLAIN BLADE, FLAT END



Shape D

TAPERED FISSURE,
PLAIN BLADE, LONG HEAD



Shape E

PEAR



Shape F

STRAIGHT FISSURE,
CROSSCUT BLADE, FLAT END



Shape G

TAPERED FISSURE, CROSSCUT BLADE



Shape H

END CUTTING

FIGURE 4 Bur head shapes

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
GG-B-00821E(DSA-DM) BURS, DENTAL EXCAVATING, TUNGSTEN CARBIDE

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
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